RIAR Consulting – Practical Assignment (Detailed)

Role: Python Backend Developer
Project: VERA – Data Intelligence Platform

Task 1: Data Ingestion Module (Backend + Frontend)

Backend Requirements

Build APIs for uploading and managing data:

1. POST /upload
   - Accept CSV/JSON file
   - Parse and store data

2. GET /datasets
   - List all uploaded datasets

3. GET /datasets/{id}
   - Fetch dataset details

Key Expectations:

- File handling (CSV/JSON parsing)
- Data validation (missing fields, wrong format)
- Store metadata (file name, upload time, record count)

Frontend Requirements

Create a simple UI (React / HTML + JS):

- File upload component
- Dataset listing page
- Dataset detail view

UI Features:

- Drag & drop upload
- Table preview of uploaded data
- Status messages (success/error)

Task 2: Data Storage & Management Layer (Backend + Frontend)

Backend Requirements

- Integrate database (PostgreSQL recommended)
- Create schema:

➢ Dataset
➢ Records (dynamic or JSON-based)

- Implement:

➢ Insert, update, delete operations
➢ Filtering & pagination APIs

APIs:

- GET /records?dataset_id=
- DELETE /dataset/{id}

Frontend Requirements

- Dashboard to manage datasets
- Features:

➢ View records in table format

➢ Pagination controls

➢ Delete dataset button

➢ Filter/search functionality

Task 3: Intelligence & Analytics Engine (Backend + Frontend)

Backend Requirements

Build a processing engine:

- Endpoint: POST /analyze/{dataset_id}

Processing Logic:

- Generate insights:
  - Total records
  - Column-wise summary
  - Trends (basic aggregation)
- Add **one intelligent feature**:
  - Anomaly detection OR
  - Rule-based classification OR
  - Keyword/entity extraction

Output Example:

```
{
  "total_records": 100,
  "columns": {
    "age": { "avg": 25, "max": 40 },
    "salary": { "avg": 50000 }
  },
  "insights": ["High salary variance detected"]
}
```

Frontend Requirements

- Analytics dashboard
- Visual representation:
  - Charts (bar/pie using Chart.js or similar)
  - Insight cards
- Button to trigger analysis

Task 4: Authentication, Security & User Management (Backend + Frontend)

Backend Requirements

- JWT-based authentication
- APIs:
  - POST /register
  - POST /login
- Role-based access:
  - Admin → full access
  - User → limited access

Security Features:

- Password hashing
- Protected routes
- Error handling

Frontend Requirements

- Login & Signup pages
- Session handling (JWT storage)
- Role-based UI:
  - Admin dashboard
  - User dashboard (restricted view)

Bonus Features (Highly Recommended)

- **Real-time updates (WebSockets)**
- **Graph-based visualization (Neo4j integration concept)**
- **Deployment (AWS / Render)**
- **Docker setup**
- **Logging system**